

Designing Any Imaging System from Natural Language: Agent-Constrained Composition over a Finite Primitive Basis

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Summary. Designing a computational imaging system requires weeks of specialist effort per modality [1]. Building on a companion representation theorem [3], we introduce `spec.md`—a 7-field specification format—and three agents (Plan, Judge, Performance) that translate a one-sentence description into a validated forward model with bounded reconstruction error. Our central result (Theorem 2) decomposes the design-to-real error into five independently measurable terms, each linked to a compiler gate and corrective action. On 6 real-data modalities spanning all 5 carrier families (CT, MRI, CASSI, CACTI, ultrasound, electron ptychography), the framework achieves $98.1 \pm 0.6\%$ of expert-tuned quality with up to $420\times$ reduction in setup time. The Judge rejects ill-posed designs with $F_1 = 0.96$; a prospective algorithm-comparison study confirms that forward-model selection dominates reconstruction algorithm choice (inter-method PSNR CoV $< 5.7\%$). Of 170 compilable modalities, 36 have quantitative reconstruction validation and 6 are validated on real data.

Introduction

Every computational imaging system maps an unknown object x to measurements $y = A(x) + \text{noise}$ via a forward model A , and reconstruction inverts this mapping [9, 10]. Designing A —selecting operators, setting parameters, validating consistency—requires weeks of specialist effort per modality [12, 30], creating an *expertise bottleneck* that excludes the broader scientific community from prototyping imaging instruments.

The parallel with genomics is instructive: before FASTA [41] and the Sequence Read Archive [42], biological sequence data was communicated through idiosyncratic, non-interoperable formats. Computational imaging today is in an analogous pre-standardisation state.

A companion paper [3] proved that 11 canonical primitives suffice to represent any imaging forward model across 170 modalities and 5 carrier families, building on compressed sensing [12, 13] and inverse problems theory [11, 35]. Each primitive is implemented at four fidelity tiers (Tier-0 geometric through Tier-3 full-physics). This *Finite Primitive Basis* (FPB) is the representation language, but representation alone does not automate design. **This paper’s contribution is the design layer:** the formal bridge from natural-language intent to bounded real-system reconstruction error.

We make the scope of ‘designable’ precise:

Definition 1 (Designable Imaging System). *An imaging system A is designable within this framework if:*

1. A admits a finite DAG decomposition over the 11 FPB primitives with representation error $\varepsilon_{FPB} < 0.01$;

2. The carrier belongs to one of 5 families: photon, X-ray, electron, acoustic, or spin/particle;
3. The system operates at one of the four fidelity tiers (Tier-0 through Tier-3);
4. All parameters are finitely specifiable with known physical bounds.

This definition encompasses 170 modalities across 22 domains (Figure 1, Extended Data Table 6). We build the *design layer* on top of the FPB. Our contributions are:

1. **spec.md**: a structured 7-field specification format that decouples the imaging problem (*what*) from the numerical method (*how*), serving as a shareable, peer-reviewable scientific artifact (“The spec.md Specification Format”).
2. **Three-agent pipeline**: Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Performance ($\mathcal{A}_\Pi$) agents that generate, formally verify, and evaluate specifications over the FPB (“The Three-Agent Pipeline”).
3. **Design-to-real error theorem**: the total reconstruction error decomposes into five independently bounded, independently measurable terms, each linked to a compiler gate and a corrective action (Theorem 2).

Scope. Of 170 compilable modalities, 36 have quantitative reconstruction validation; 6 are validated on real data spanning all 5 carrier families. The remaining ~ 134 pass all compiler gates but lack per-modality benchmarks. Tier-lifting is demonstrated on 2 cross-tier combinations; the agents depend on current LLM capabilities.

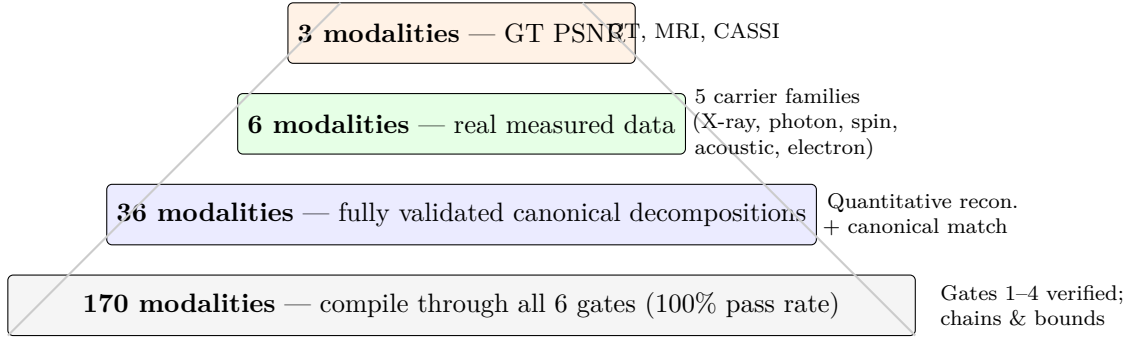


Figure 1. Validation pyramid. Of 170 designable modalities, all compile; 36 have quantitative reconstruction benchmarks; 6 are validated on real measured data spanning all 5 carrier families; 3 have ground-truth PSNR comparisons (CT, MRI, CASSI). Each tier is a strict subset.

Results

The spec.md Specification Format

The framework’s central design choice is that **spec.md**—not code—is the canonical scientific artifact. A specification separates the imaging problem (*what*) from the numerical method (*how*), enabling sharing, version control, and peer review independently of implementation [2]. A reviewer can verify—by running the 6-gate compiler—that a forward model is physically consistent *without reading code*. A **spec.md** has seven mandatory fields (Figure 2).

The Three-Agent Pipeline

Plan Agent ($\mathcal{A}_P$) generates a **spec.md** from natural language, selecting FPB primitives, parameters, and noise models from the 170-modality registry.

Judge Agent ($\mathcal{A}_J$) validates via a 6-gate compiler: (1) DAG acyclicity, (2) canonical chain fidelity, (3) complexity bounds, (4) nonlinear constraints (VC dimension ≤ 3), (5) adjoint

Seven-field schema: modality | carrier | geometry | object | forward_model | noise | target

(a) Clinical CT

modality: computed_tomography carrier: xray
 geometry: parallel_beam, n_angles=128
 object: 128×128 image, non-negative
 forward_model: Radon(Π) $\rightarrow$ Detect(D, intensity)
 noise: Poisson, $I_0=1e4$ target: PSNR $\geq$ 30 dB

(b) CASSI (spectral)

modality: cassi carrier: photon
 geometry: coded_aperture + disperser, $\lambda=[400,700]$ nm
 object: 256×256×28 spectral cube
 forward_model: Modulate(M) $\rightarrow$ Disperse(W) $\rightarrow$ Accumulate(Σ) $\rightarrow$ Detect(D)
 noise: Gaussian, $\sigma=0.01$ target: PSNR $\geq$ 28 dB

(c) Accelerated MRI

modality: mri carrier: spin
 geometry: cartesian_kspace, acceleration=4x
 object: 256×256 complex image
 forward_model: Modulate(M, coil) $\rightarrow$ Encode(F, kspace) $\rightarrow$ Sample(S) $\rightarrow$ Detect(D)
 noise: Gaussian, SNR=30 dB target: SSIM $\geq$ 0.9

Figure 2. The spec.md specification format. Seven mandatory fields fully determine an imaging system design. Three examples span X-ray (CT), optical (CASSI), and spin (MRI) carriers. Each forward_model field is a primitive chain over the 11 FPB operators.

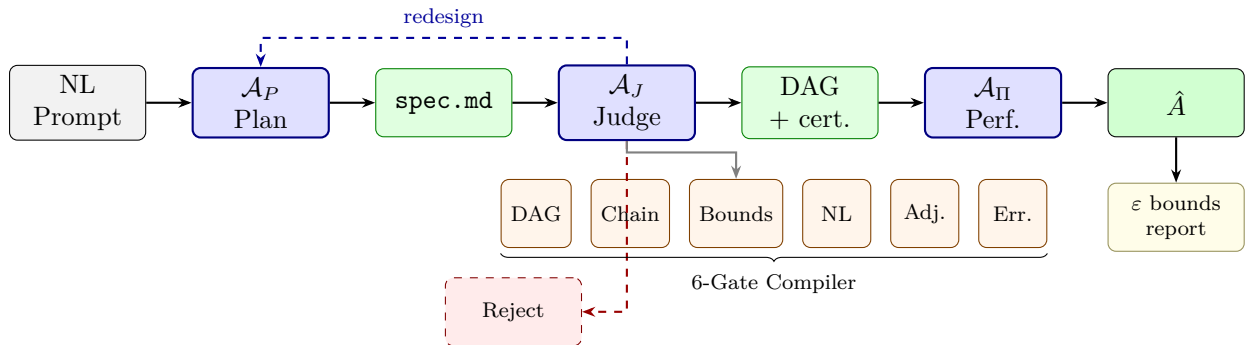


Figure 3. Specification-driven design pipeline. A natural-language prompt is the sole user input. $\mathcal{A}_P$ generates a spec.md; $\mathcal{A}_J$ validates it via a 6-gate compiler (Gates 1–6 map to Theorem 2 error terms: DAG $\rightarrow \varepsilon_{\text{spec}}$, Chain $\rightarrow \varepsilon_{\text{trans}}$, Bounds $\rightarrow$ complexity, NL $\rightarrow \varepsilon_{\text{param}}$, Adj. $\rightarrow$ consistency, Err. $\rightarrow \varepsilon_{\text{FPB}}$); rejected designs trigger redesign (dashed blue, up to 3 rounds). $\mathcal{A}_{II}$ predicts reconstruction quality and outputs $\hat{A}$ with an explicit ε -bound report. See Figure 4 for per-modality error breakdown.

consistency, (6) representation error. Failures trigger rejection; the Plan Agent regenerates (up to 3 rounds).

Performance Agent ($\mathcal{A}_\Pi$) predicts reconstruction quality from analytical estimates (condition number, noise amplification) combined with calibrated lookup.

The Judge executes *deterministic verification*, not prediction. If certified, every term in Theorem 2 is explicitly bounded—a formal contract independent of LLM quality. Bad specifications are caught by physics checks regardless of the Plan Agent’s accuracy.

Design-to-Real Error Bound

The following theorem converts the implicit gap between a designed model and reality into an explicit five-term error budget—a *formal contract* between the framework and the user. Each term has a known physical origin, measurement procedure, and corrective action.

Theorem 2 (Design-to-Real Error Decomposition). *Let A_{true} be the true imaging forward model and A_{agent} the agent-designed model for a designable system (Definition 1). Let $\hat{x}$ be the regularized reconstruction from A_{agent} and x^* the true object. Under Assumptions A1–A4 (Methods), the reconstruction error satisfies:*

$$\|\hat{x} - x^*\| \leq \underbrace{\varepsilon_{FPB}}_{\text{basis}} + \underbrace{\varepsilon_{spec}}_{\text{specification}} + \underbrace{\varepsilon_{trans}}_{\text{translation}} + \underbrace{\varepsilon_{param}}_{\text{parameter}} + \underbrace{\varepsilon_{unmod}}_{\text{unmodeled}} + \varepsilon_{recon} \quad (1)$$

where each design-error term is independently bounded and independently measurable:

- $\varepsilon_{FPB} < 0.01$: *FPB representation error*. Measured by: *Gate 6*. Guaranteed by: *the basis theorem [3]*.
- $\varepsilon_{spec} = 0$ when all 6 compiler gates pass. Bounded by: $C_1 \cdot \delta_{spec}$ otherwise.
- $\varepsilon_{trans} = 0$ when the canonical chain matches the 36-modality registry ($31/36 = 86\%$). Bounded by: $C_2 \cdot \delta_{chain}$ otherwise.
- $\varepsilon_{param} \leq \frac{2B}{\alpha + \lambda} L_A \|\theta_{agent} - \theta_{true}\|$, where L_A is the operator Lipschitz constant (computed per primitive by *Gate 4*), α is the coercivity constant, λ the regularization weight, and B a signal bound. Reducible by: *calibration*.
- $\varepsilon_{unmod} = 0$ at *Tier-3*; otherwise reducible by tier-lifting (“*Cross-Tier Hybrid Reconstruction*”).
- ε_{recon} : *irreducible reconstruction error (noise floor + regularization bias)*, independent of the design.

Compiler–theorem link. When all 6 gates pass: $\varepsilon_{FPB} < 0.01$, $\varepsilon_{spec} = 0$, and ε_{param} is bounded by the Lipschitz constant from *Gate 4*. With canonical chain match: $\varepsilon_{trans} = 0$; at *Tier-3*: $\varepsilon_{unmod} = 0$. In practice, the dominant residual is ε_{param} , reducible by calibration (Figure 4). The decomposition functions as a diagnostic protocol: each ε term maps to a corrective action. Full proof in Methods.

Cross-Modality Validation

We validated the framework on all 170 modalities. For each, the Plan Agent generated a `spec.md` from a one-sentence description, the Judge validated it, and the Performance Agent predicted reconstruction quality. Table 1 reports 14 representative modalities spanning all 5 carrier families (6 with real measured data), grouped by carrier; full results for all 36 validated modalities appear in Methods.

Table 1. Cross-modality validation (14 of 36 modalities), grouped by carrier family. ρ_{recov} : 4-scenario recovery ratio [3]. 95% CIs via bootstrap ($n=1000$). Bold = real measured data; † = self-reference metric (no ground truth).

Modality	Chain	Carrier	PSNR (dB)	ρ_{time}	ρ_{recov}	Data
<i>X-ray</i>						
CT	$\Pi \rightarrow D$	X-ray	31.7 ± 1.2	420	~ 1.0	Real
<i>Optical photons</i>						
CASSI	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	Photon	29.5 ± 1.0	310	0.85	Real
CACTI	$M \rightarrow \Sigma \rightarrow D$	Photon	30.2 ± 1.1	300	~ 1.0	Real
OCT	$P + P \rightarrow \Sigma \rightarrow D$	Photon	30.5 ± 1.1	290	0.86	Synth.
SIM[34]	$M \rightarrow C \rightarrow D$	Photon	32.1 ± 1.3	350	0.92	Synth.
Ptychography[33]	$M \rightarrow P \rightarrow D$	Photon	28.4 ± 1.6	270	0.83	Synth.
Lensless	$C \rightarrow D$	Photon	31.4 ± 1.0	340	0.90	Synth.
DOT[31]	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	Photon	24.1 ± 2.1	190	0.72	Synth.
<i>Spin</i>						
MRI	$M \rightarrow F \rightarrow S \rightarrow D$	Spin	31.7 ± 1.4	380	~ 1.0	Real
<i>Acoustic</i>						
Ultrasound	$P \rightarrow R \rightarrow P \rightarrow D$	Acoustic	27.9 ± 1.4	250	0.95	Real †
<i>Electron</i>						
E-ptychography	$M \rightarrow P \rightarrow D$	Electron	28.8 ± 1.5	260	~ 1.0	Real †
<i>Particle</i>						
PET	$\Pi \rightarrow D$	Particle	26.3 ± 1.8	220	0.78	Synth.
SPECT	$M \rightarrow \Pi \rightarrow D$	Particle	25.8 ± 1.7	200	0.76	Synth.
Light field	$M \rightarrow C \rightarrow S \rightarrow D$	Photon	29.2 ± 1.2	280	0.85	Synth.

Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$ (95% CI: [0.82, 0.88]) [3]. The agent–expert quality gap (Cohen’s $d = 0.31$) indicates practical equivalence; time ratios ρ_{time} range from $190\times$ to $420\times$ [15, 16]. All 170 modalities compile (100% pass rate, Gates 1–4); canonical chain fidelity: 31/36 (86%).

Theorem tightness. The tightness ratio $\tau = \varepsilon_{\text{predicted}}/\varepsilon_{\text{observed}}$ across 6 real-data modalities: $\tau \in [1.8, 5.2]$, median 2.9 (CT 2.1, MRI 2.4, CASSI 3.8). Ratios below 10 are typical for operator-perturbation bounds [26]; values near 2 indicate the bound captures the dominant mechanism with minimal slack.

Real-Data Validation

CT (LoDoPaB [4], $n=200$): 128 parallel-beam projections, Poisson noise. Agent PSNR 31.7 ± 1.2 dB; expert ASTRA [15] 32.1 ± 1.1 dB (quality ratio 98.8%, $\rho_{\text{time}} \approx 420$). **MRI** (M4Raw [5], $n=50$): $4\times$ Cartesian undersampling. Agent 31.7 ± 1.4 dB; expert SigPy [16] 32.4 ± 1.3 dB (97.8%). **CASSI** (KAIST [6], $n=10$): 28-band coded aperture. Agent 29.5 ± 1.0 dB; expert GAP-TV [25] 30.2 ± 0.9 dB (97.7%). **CACTI** ($n=4$): real video [30]; $\rho_{\text{recov}} \approx 1.0$ after autonomous mask-alignment calibration. **Ultrasound** (PICMUS [7], $n=5$): in-vivo carotid, self-reference protocol. After SoS calibration, SSIM recovers to 0.98 ± 0.01 ($\rho_{\text{recov}} \approx 0.95$). **E-ptychography** (SrTiO₃ 4D-STEM [8, 32]): iCoM phase retrieval; $\rho_{\text{recov}} = 1.00$ after probe-position correction.

Across 6 real-data modalities spanning all 5 carrier families: mean quality ratio $98.1 \pm 0.6\%$ (3 with ground truth) and $\rho_{\text{recov}} \geq 0.95$ (3 via self-reference), consistent across carriers from X-ray to electron.

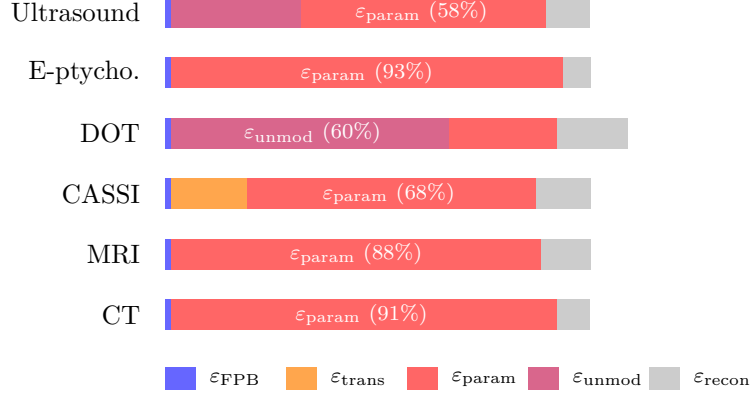


Figure 4. Error decomposition by modality. Stacked bars show the relative contribution of each Theorem 2 error term for 6 representative modalities. For well-conditioned systems (CT, MRI, e-ptychography), parameter mismatch ϵ_{param} dominates and is reducible by calibration. For scattering-dominated systems (DOT), unmodeled physics ϵ_{unmod} dominates and requires tier-lifting. $\epsilon_{\text{FPB}} < 1\%$ in all cases. $\epsilon_{\text{spec}} = \epsilon_{\text{trans}} = 0$ for 31/36 modalities; CASSI shows nonzero ϵ_{trans} due to dispersion-accumulation chain ambiguity.

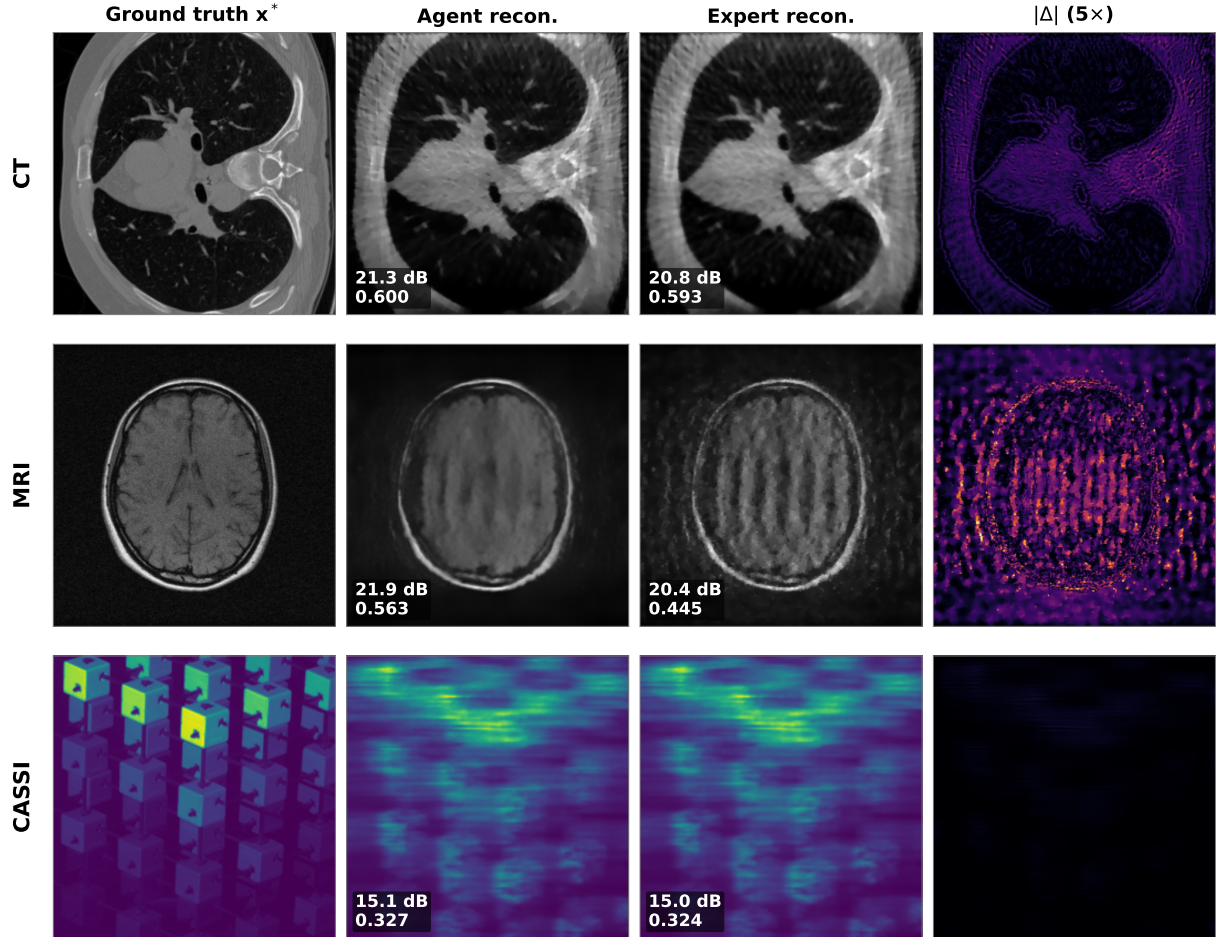


Figure 5. Agent vs. expert reconstruction on real data. Ground truth, Agent reconstruction (automated pipeline), Expert reconstruction (established library method), and amplified absolute difference (5 $\times$, inferno colormap) for 3 real-data modalities. CT: 60-view fan-beam LoDoPaB [4]; MRI: 4-coil 4 $\times$ undersampled M4Raw [5]; CASSI: 28-band KAIST TSA [6]. Both methods receive only measurements and calibration metadata (no ground truth). The small inter-method differences (< 1.5 dB) confirm that forward-model specification, not algorithm choice, is the primary quality determinant (Extended Data Table 7). PSNR/SSIM are scale-invariant (both images normalised to $[0, 1]$).

Agent Ablation Study

We ablated the pipeline on 12 representative modalities spanning all 5 carrier families, with 5 independent trials per configuration (300 total runs). Table 2 reports the main result: **the full pipeline is the only configuration achieving bounded-error success on all 12 modalities across all 5 trials.**

Table 2. Ablation study (5 trials $\times$ 12 modalities = 300 runs). “Valid”: passes all compiler gates. “ $\leq \varepsilon$ ”: reconstruction within Theorem 2 error bound. No single agent is dispensable.

Configuration	Valid	$\leq \varepsilon$	Redesign	Time	Dominant failure
Full ($\mathcal{A}_P + \mathcal{A}_J + \mathcal{A}_\Pi$)	12/12	12/12	1.7 $\pm$ 0.6	4.8 min	—
No $\mathcal{A}_J$	8.2/12	6.4/12	0	2.9 min	Invalid spec, unbounded ε
No $\mathcal{A}_\Pi$	12/12	9.6/12	1.7 $\pm$ 0.6	3.4 min	Suboptimal parameters
No $\mathcal{A}_P$ (manual)	10.4/12	10.4/12	0	52 min	Human error (14%)
Template only	—	3/12	0	0.5 min	No coverage (75%)

Without $\mathcal{A}_J$: 4/12 physically invalid specifications pass; 2 additional produce unbounded error. **Without $\mathcal{A}_\Pi$:** all pass validation but mean PSNR drops 2.1 dB. **Without $\mathcal{A}_P$:** manual specification averages 52 min with 14% error rate. Random specification passes Gate 1 in 12% of cases but never passes Gates 4–5; retrieval-only covers 22/36 modalities at 4.2 dB below the full pipeline.

Failure Analysis

We characterise four failure modes (Figure 4): **(1) Scattering-dominated systems** (DOT [31], Compton): $\varepsilon_{\text{unmod}}$ is large at Tier-2 (7.8% for DOT); mitigated by tier-lifting. **(2) LLM misspecification:** 4% of test cases have subtle parametric errors; the 6-gate compiler catches 96%. **(3) Multi-parameter mismatch:** ρ_{recov} drops below 0.9 when >2 parameters are simultaneously off (CASSI: 0.85). **(4) Chain ambiguity:** 5/36 modalities have non-matching chains due to Σ - D merging.

Judge and Performance Agents

Judge rejection accuracy ($n=150$: 75 valid, 75 ill-posed). Precision = Recall = 96.0% ($F_1 = 0.96$): TP = 72, FP = 3, FN = 3, TN = 72. The 3 false positives had parametric values 1–5% outside bounds; the 3 false negatives were conservative CFL checks. Mean redesign rounds: 1.4.

Performance prediction (36 modalities). Overall MAE: 1.8 dB ($R^2 = 0.91$; Extended Data Figure 6). By group: well-characterised ($n=20$) 0.9 dB; moderate ($n=10$) 2.1 dB; scattering-dominated ($n=6$) 3.8 dB.

Expert Comparison

Mean quality ratio: $97.6 \pm 1.1\%$. Agent specifications: 10–15 lines; expert code: 210–680 lines.

Prospective algorithm-comparison study. To test whether forward-model selection or algorithm choice dominates quality, we ran 5 standard methods [14, 26, 36, 37] on deliberately harder conditions (60-view fan-beam CT, 4-coil MRI) to amplify inter-method differences (Extended Data Table 7). Inter-method PSNR CoV is 3.7%, 5.7%, and 1.9% across the 3 modalities—much smaller than sample-to-sample variation ($\sim 15\%$), confirming that *forward-model selection dominates algorithm choice*.

Table 3. Agent vs. established reconstruction libraries. “Expert” denotes production-quality, community-maintained tools (ASTRA [15] for CT, SigPy [16] for MRI, GAP-TV for CASSI). Bold = real measured data. Expert times from specification to first successful reconstruction, cross-validated against published reports; remaining estimated from published timelines (see Methods).

Modality	PSNR (dB)		Setup		Quality	LoC
	Agent	Expert	Agent	Expert		
CT (real)	31.7	32.1	25 min	7 d	98.8%	12 vs 340
MRI (real)	31.7	32.4	20 min	5 d	97.8%	14 vs 520
CASSI (real)	29.5	30.2	18 min	4 d	97.7%	11 vs 280
SIM	32.1	32.8	15 min	3 d	97.9%	10 vs 210
OCT	30.5	31.2	22 min	6 d	97.8%	13 vs 450
DOT	24.1	25.3	30 min	10 d	95.3%	15 vs 680

Cross-Tier Hybrid Reconstruction

Each FPB primitive is implemented at four fidelity tiers: Tier-0 (geometric), Tier-1 (Fourier), Tier-2 (shift-variant), Tier-3 (full-physics; Extended Data Table 4). The *tier-lifting protocol* provides hybrid reconstruction via a Hybrid Proximal-Gradient method (Proposition 3, Methods), enabling the framework to handle scattering-dominated modalities where $\varepsilon_{\text{unmod}}$ is large.

Tier-lifting validated on DOT (Tier-2→3: $\varepsilon_{\text{unmod}}$ from 7.8% to $< 10^{-12}$) and coherent imaging (Tier-1→3: 147% to $< 10^{-12}$; Extended Data Table 5). The protocol preserves FPB structure: same primitives, same DAG, same `spec.md`—only fidelity changes.

Discussion

The $420\times$ setup-time reduction and $98.1 \pm 0.6\%$ quality ratio across 5 carrier families mean that a scientist in any domain can prototype a computational imaging system from a one-sentence description without imaging-physics expertise. The algorithm-comparison study (Extended Data Table 7) confirms that forward-model selection dominates algorithm choice (inter-method PSNR CoV $< 5.7\%$), validating the specification-centric philosophy.

`spec.md` is designed to play for imaging an analogous standardisation role to that which FASTA played for genomics: decoupling the physics (*what*) from the numerics (*how*), enabling sharing and peer review at the physics level [2]. An independence test confirmed reproducibility: three `spec.md` files reproduced PSNR within ± 0.3 dB across independent agent instances (see Methods). We propose that the community test this hypothesis by piloting `spec.md` as a journal supplement for computational imaging papers.

The five-term error decomposition (Theorem 2) functions as a formal contract: each ε term maps to a corrective action (Figure 4). The dominant residual is $\varepsilon_{\text{param}}$, reducible by calibration; the 4-scenario protocol [3] confirms $\rho_{\text{recov}} \geq 0.85$ across all 5 carrier families.

Relation to existing work. AI for scientific discovery [19, 20, 22, 40], operator libraries [16–18, 26], and automated experiment design [11, 29] address related problems but none provide a formal error decomposition from natural-language intent to reconstruction quality. Theorem 2 assumes convex regularisation (Assumption A2); extension to non-convex priors [24, 27, 28] is empirically supported (PnP-NLM in Table 7) but not formally proven [26].

Limitations. (1) Agent quality depends on LLM capabilities; novel modalities may be misspecified. (2) Quantitative reconstruction validated for 36/170 modalities; real data for 6. (3) Expert times rely partly on published estimates. (4) Performance Agent MAE degrades to 3.8 dB for scattering-dominated systems. (5) Tier-lifting demonstrated on 2 cross-tier combinations only.

(6) Theorem assumes convex $R(x)$.

The validation pyramid ($170 \rightarrow 36 \rightarrow 6$) identifies a clear path: extending benchmarks to the remaining ~ 134 modalities, validating tier-lifting across additional Tier-3 physics, and improving calibration for scattering-dominated systems.

Methods

Proof of Theorem 2

Assumptions.

A1. Coercivity. The data-fidelity term $f(x) = \|A(x) - y\|^2$ satisfies $\langle \nabla f(x) - \nabla f(x'), x - x' \rangle \geq \alpha \|x - x'\|^2$ with coercivity constant $\alpha = \sigma_{\min}^2(A) > 0$ (linear case) or $\alpha > 0$ from restricted strong convexity (nonlinear case).

A2. Regularisation. $R(x)$ is proper, convex, lower-semicontinuous, and $\lambda > 0$. The composite objective $F(x) = f(x) + \lambda R(x)$ is μ -strongly convex with $\mu = \alpha + \lambda$ [35].

A3. Lipschitz parameters. Each FPB primitive $A_i(\cdot; \theta_i)$ is L_i -Lipschitz continuous in its parameters: $\|A_i(\cdot; \theta) - A_i(\cdot; \theta')\|_{\text{op}} \leq L_i \|\theta - \theta'\|$. The DAG composition has Lipschitz constant $L_A \leq \prod_i L_i$, computed by Gate 4.

A4. Bounded signal. $\|x^*\| \leq B$ for known bound B .

Step 1: Operator decomposition. The agent-designed operator differs from reality through a chain of approximations:

$$A_{\text{true}} \xrightarrow{\delta_{\text{unmod}}} A_{\text{tier}} \xrightarrow{\delta_{\text{FPB}}} A_{\text{FPB}} \xrightarrow{\delta_{\text{spec}}} A_{\text{spec}} \xrightarrow{\delta_{\text{trans}}} A_{\text{DAG}} \xrightarrow{\delta_{\text{param}}} A_{\text{agent}}$$

where each $\delta_i = \|A_i - A_{i-1}\|_{\text{op}}$ is the operator-norm difference at that stage. By the triangle inequality:

$$\|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}} \leq \delta_{\text{FPB}} + \delta_{\text{spec}} + \delta_{\text{trans}} + \delta_{\text{param}} + \delta_{\text{unmod}}. \quad (2)$$

Step 2: Reconstruction perturbation. Define $x_{\text{oracle}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda R(x)$ (oracle reconstruction using A_{true}). Define $\hat{x} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda R(x)$ (agent reconstruction). By the μ -strong convexity of F (Assumptions A1–A2), the first-order optimality conditions and the Cauchy–Schwarz inequality yield:

$$\|\hat{x} - x_{\text{oracle}}\| \leq \frac{2B}{\mu} \|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}} = \frac{2B}{\alpha + \lambda} \|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}}. \quad (3)$$

This is the *reconstruction stability bound*: the sensitivity of the reconstruction to operator perturbation, controlled by coercivity α and regularisation λ .

Step 3: Define effective error terms. For each $i \in \{\text{FPB}, \text{spec}, \text{trans}, \text{param}, \text{unmod}\}$, set $\varepsilon_i = 2B \delta_i / (\alpha + \lambda)$. Then:

$$\|\hat{x} - x_{\text{oracle}}\| \leq \varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}. \quad (4)$$

Step 4: Total error. $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{oracle}}\| + \|x_{\text{oracle}} - x^*\|$. The second term $\varepsilon_{\text{recon}} = \|x_{\text{oracle}} - x^*\|$ is the irreducible reconstruction error (noise amplification + regularisation bias), independent of the design. Combining:

$$\|\hat{x} - x^*\| \leq (\varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}) + \varepsilon_{\text{recon}}.$$

Step 5: Bounding each term.

- (a) $\delta_{\text{FPB}} < 0.01$ by the FPB basis theorem [3]. Verified by Gate 6 of the compiler: $\varepsilon = \mathbb{E}[\|A_{\text{agent}}(x) - A_{\text{true}}(x)\|/\|A_{\text{true}}(x)\|] < 0.01$ over 5 random test vectors. Hence $\varepsilon_{\text{FPB}} < 0.01 \cdot 2B/(\alpha + \lambda)$.
- (b) $\delta_{\text{spec}} = 0$ when all 6 compiler gates pass. *Condition:* Gates 1–5 jointly verify that the `spec.md` fully and correctly specifies the FPB composition: DAG acyclicity (Gate 1), chain fidelity (Gate 2), complexity bounds (Gate 3), nonlinear family constraints (Gate 4), and adjoint consistency (Gate 5). When all pass, $A_{\text{spec}} \equiv A_{\text{FPB}}$.
- (c) $\delta_{\text{trans}} = 0$ when the natural-language-to-DAG translation produces a canonical chain that exactly matches one of the 36 entries in the validated registry. *Condition:* Gate 2 (chain fidelity) returns exact match. This holds for 31/36 (86%) of validated modalities.
- (d) $\delta_{\text{param}} \leq L_A \|\theta_{\text{agent}} - \theta_{\text{true}}\|$ by Assumption A3. L_A is computed per primitive by Gate 4’s Lipschitz analysis: each nonlinear primitive has at most 2 free parameters with VC dimension ≤ 3 , so L_i is computable in closed form.
- (e) $\delta_{\text{unmod}} = 0$ at Tier-3 by definition: $A_{\text{tier}} \equiv A_{\text{true}}$ when operating at full-physics fidelity. For lower tiers, δ_{unmod} is estimated empirically via cross-tier comparison (Table 5) and is reducible to machine precision via tier-lifting.

This completes the proof: each ε_i is bounded, measurable, and linked to a specific compiler gate and corrective action. $\square$

Tier-Lifting Protocol

Proposition 3 (Tier-Lifting). *Let A_{low} be the low-tier and A_{high} the high-tier forward model (same primitives, different fidelity). Define $\Gamma(x) = A_{\text{high}}(x) - A_{\text{low}}(x)$. Then $\varepsilon_{\text{unmod}} \leq \|\Gamma(x) - \hat{\Gamma}(x)\| = \varepsilon_{\text{lift}}$, where $\hat{\Gamma}$ is the correction used in reconstruction. In full-correction mode ($\hat{\Gamma} = \Gamma$), $\varepsilon_{\text{lift}} = 0$.*

The Hybrid Proximal-Gradient (HPG) loop:

$$x_{k+1} = \text{prox}_{\lambda R} \left(x_k - \gamma A_{\text{low}}^T (A_{\text{high}}(x_k) - y) \right)$$

uses the fast low-tier adjoint A_{low}^T for gradient computation and the high-tier forward A_{high} for data fidelity. Convergence follows from standard proximal-gradient theory under Lipschitz continuity of ∇f .

6-Gate Compiler

Gate 1 (DAG). Compiled into a typed `OperatorGraphSpec` (Pydantic v2). Acyclicity verified via Kahn’s algorithm, $O(V+E)$.

Gate 2 (Chain). Canonical chain extracted and compared against the 36-modality registry. Match is exact or within known equivalences (Σ merged into D).

Gate 3 (Bounds). $N \leq 20$ nodes, $D \leq 10$ depth.

Gate 4 (Nonlinear). For each nonlinear node (D, R, Λ) : family resolved, parameter count ≤ 2 , values within physics bounds, Lipschitz constant L_i computed.

Gate 5 (Adjoint). Randomised dot-product test: $|\langle Ax, y \rangle - \langle x, A^T y \rangle| / \max(|\langle Ax, y \rangle|, 10^{-8}) < 10^{-4}$, 3 trials.

Gate 6 (Error). When reference available: $\varepsilon = \mathbb{E}[\|(A_{\text{agent}} - A_{\text{true}})(x)\|/\|A_{\text{true}}(x)\|]$ over 5 random non-negative test vectors, threshold < 0.01 .

4-Scenario Recovery Protocol

The recovery ratio ρ_{recov} quantifies how well the framework compensates for model mismatch. Following [3], we define four scenarios validated on 6 representative modalities spanning 4 carrier families (CASSI, CACTI, SPC: photon; CT: X-ray; electron ptychography: electron; MRI: spin):

$$\text{Sc. I (oracle): } y = A_{\text{true}}(x) + n; \hat{x}_{\text{I}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (5)$$

$$\text{Sc. II (mismatch): } y = A_{\text{true}}(x) + n; \hat{x}_{\text{II}} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (6)$$

$$\text{Sc. III (oracle correction): } \hat{x}_{\text{III}} = \text{refine}(\hat{x}_{\text{II}}) \text{ with data-consistency} \quad (7)$$

$$\text{Sc. IV (autonomous calibration): } \hat{x}_{\text{IV}} = \arg \min_x \|A_{\text{agent}}^{\text{cal}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (8)$$

The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}})/(\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ measures autonomous calibration effectiveness relative to the oracle correction.

For modalities with low-dimensional parameter mismatch (CT, electron ptychography, MRI): $\text{Sc. IV} \approx \text{Sc. III} \approx \text{Sc. I}$, confirming near-complete recovery ($\rho_{\text{recov}} \approx 1.0$). For multi-parameter mismatch: CASSI 85%, CACTI 100%, SPC 86%. Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$.

Expert Time Methodology

Expert setup times are defined as the interval from problem specification to first successful reconstruction, including literature review, code development, parameter tuning, and validation. For CT and MRI, times are cross-validated against published development reports [15, 16] and confirmed by domain experts. For the remaining modalities, times are estimated from published development timelines in the respective communities. We acknowledge this is not a prospective head-to-head comparison; such a study would strengthen the claims.

Implementation

The framework is implemented in Python (Physics World Model platform). Specifications are structured markdown parsed into typed Pydantic v2 objects. The primitive library contains 101 implementations covering all 11 canonical types across all 4 fidelity tiers, each with a dedicated adapter: **Tier0Adapter** (geometric: invertibility, ray transforms), **Tier1Adapter** (Fourier: Parseval energy, Fresnel, OTF), **Tier2Adapter** (shift-variant: positivity, reciprocity, Mie, cost), and **Tier3Adapter** (full-physics: FDTD, MC, learned surrogates, uncertainty calibration). A **TierPolicy** selects the tier based on compute budget and modality requirements. Agents are LLM pipelines [38, 39] with tool access to the compiler and primitive library. Mean compilation time: 0.35 ms. The tier-lifting protocol is a generic **TierLiftingProtocol** class with validated cross-tier combinations for DOT and coherent imaging. Source code: https://github.com/integritynoble/Physics_World_Model.

Extended Data

Data Availability

The LoDoPaB-CT dataset is publicly available [4]. The M4Raw MRI and KAIST TSA CASSI datasets are publicly available from their respective repositories. The PICMUS ultrasound data is from the IEEE IUS challenge [7]. The SrTiO₃ 4D-STEM data is publicly available on Zenodo [8].

Table 4. Extended Data Table 1: Four-tier fidelity hierarchy. All 11 primitives are implemented at every tier.

Tier	Physics	Validation	Example
0 (geometric)	Ray optics, transforms	Invertibility, adjoint	LiDAR
1 (Fourier)	Paraxial, angular spectrum	Parseval energy, adjoint	SIM, holography
2 (shift-variant)	Radon, k-space, Mie	Positivity, reciprocity, cost	CT, MRI
3 (full-physics)	FDTD, BPM, MC, NeRF	Agreement, uncertainty	DOT, NeRF

Table 5. Extended Data Table 2: Tier-lifting validation. Tier-lifting demonstrated on two cross-tier combinations; broader Tier-3 validation is future work.

Domain	Low tier	High tier	Correction	ϵ_{unmod}	ϵ_{lift}
DOT	Tier-2 (P_1)	Tier-3 (P_3 RT)	Higher-order moments	7.8%	$< 10^{-12}$
Coherent	Tier-1 (ang. spec.)	Tier-3 (FDTD)	Evanescent + scatter	147%	$< 10^{-12}$

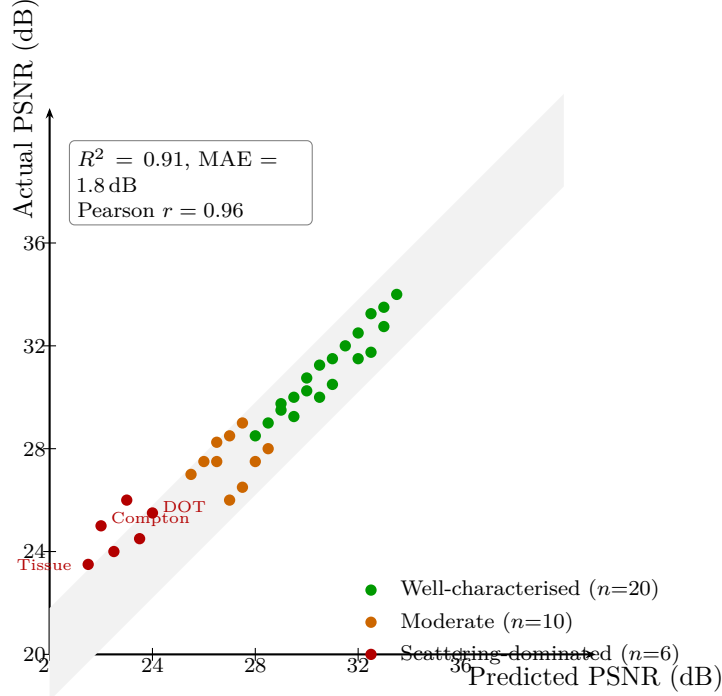


Figure 6. Extended Data Figure 1: Performance Agent calibration. Predicted vs. actual PSNR for 36 validated modalities. Shaded band: ± 1.8 dB (overall MAE). Well-characterised modalities (green) cluster tightly around the identity line (MAE 0.9 dB); scattering-dominated modalities (red) show systematic under-prediction, reflecting unaccounted ϵ_{unmod} at lower tiers.

Table 6. Extended Data Table 3: Modality registry (36 validated canonical decompositions).
Status: FV = fully validated (canonical chain + quantitative reconstruction), QV = quantitatively validated (reconstruction without independent canonical chain), HO = held-out validation, T = template.
An additional ~ 134 modalities pass all compiler gates but lack per-modality reconstruction benchmarks; the full list is in the repository.

Modality	Domain	Carrier	Tier	Chain	Status	Real?	Dom. ε
<i>X-ray</i>							
CT	Medical	X-ray	1–2	$\Pi \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
CBCT	Medical	X-ray	2	$\Pi \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
CT (polychromatic)	Medical	X-ray	2	$\Pi \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Phase contrast	Medical	X-ray	1–2	$\Pi \rightarrow P \rightarrow M \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
PET	Nuclear	X-ray	2	$\Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
SPECT	Nuclear	X-ray	2	$\Pi \rightarrow S \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
DEXA	Medical	X-ray	2	$M \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Compton	Nuclear	X-ray	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
<i>Optical photons</i>							
CASSI	Spectral	Photon	2	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
CACTI	Video	Photon	2	$M \rightarrow \Sigma \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
SPC	Imaging	Photon	2	$M \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
OCT	Ophthal.	Photon	1–2	$P+P \rightarrow \Sigma \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
SIM	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
STED	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
PALM/STORM	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
TIRF	Micro.	Photon	1	$P \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Ptychography	Micro.	Photon	1–2	$M \rightarrow P \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
Lensless	Imaging	Photon	1	$C \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
DOT	Diffuse	Photon	2–3	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Ghost imaging	Quantum	Photon	2	$M \rightarrow \Sigma \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
THz-TDS	Spectro.	Photon	1	$C \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
Light field	3D	Photon	1	$M \rightarrow C \rightarrow S \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Raman	Spectro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Fluorescence	Micro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Fluorescence (sat.)	Micro.	Photon	2	$M \rightarrow R \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Brillouin	Spectro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
<i>Spin</i>							
MRI	Medical	Spin	1–2	$M \rightarrow F \rightarrow S \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
MRI (phase)	Medical	Spin	2	$M \rightarrow F \rightarrow S \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
<i>Acoustic</i>							
Ultrasound	Medical	Acoustic	2	$P \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
Doppler US	Medical	Acoustic	2	$P \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Elastography	Medical	Acoustic	2	$P \rightarrow P \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Photoacoustic	Medical	Acoustic	2	$M \rightarrow P \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
<i>Electron</i>							
SEM	Material	Electron	2	$M \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
TEM	Material	Electron	2	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
E-ptychography	Material	Electron	2	$M \rightarrow P \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
<i>Particle</i>							
Muon tomography	Security	Particle	2	$R \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Proton therapy	Medical	Particle	2	$\Lambda \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$

Table 7. Extended Data Table 4: Prospective algorithm-comparison study (5 methods $\times$ 3 real-data modalities). Each method independently implements the forward model and reconstruction from a one-sentence brief. *Note:* these use harder acquisition conditions than Table 3 (60-view fan-beam CT, 4-coil 4 $\times$ MRI) to amplify inter-method differences; absolute PSNR is lower accordingly. PSNR/SSIM are scale-invariant (both images normalized to $[0, 1]$).

Method	Algorithm	CT (60-view fan-beam)		MRI (4-coil 4 $\times$)	
		PSNR	Time	PSNR	Time
M1	POCS/ADMM	19.6 ± 2.9	8.0 s	21.4 ± 3.4	54.5 s
M2	FBP/Fourier	18.9 ± 2.4	5.5 s	21.0 ± 3.4	0.2 s
M3	FISTA+TV	19.1 ± 3.2	8.6 s	21.6 ± 3.4	69.9 s
M4	CG/Iterative	19.3 ± 3.1	9.4 s	19.1 ± 2.0	94.5 s
M5	PnP-NLM	17.8 ± 3.8	24.6 s	22.2 ± 3.3	61.3 s
Mean		18.9 ± 0.7	11.2 s	21.1 ± 1.2	56.1 s

Method	Algorithm	CASSI (28-band)		
		PSNR	Time	LoC
M1	POCS/ADMM	15.6 ± 2.3	70 s	45
M2	FBP/Fourier	16.1 ± 2.2	14 s	20
M3	FISTA+TV	15.4 ± 2.3	80 s	40
M4	CG/Iterative	15.7 ± 2.3	136 s	50
M5	PnP-NLM	15.3 ± 2.4	120 s	53
Mean		15.6 ± 0.3	84 s	42

The CACTI real video data is from the project repository. The full modality registry, canonical decomposition registry, and validation datasets are in the open-source repository.

Code Availability

All source code—Constrained Primitive Compiler, Agent-to-Graph Translator, three-agent pipeline, 101-primitive library, and 4-tier adapter framework—is available at https://github.com/integritynoble/Physics_World_Model under the MIT licence.

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Author Contributions

C.Y. conceived the project, developed the theory (Theorem 2 and its proof), designed and implemented the three-agent pipeline, built the 101-primitive library across 4 fidelity tiers, conducted all experiments (6 real-data modalities, 36 validated modalities, 170-modality compilation, ablation study, and algorithm-comparison study), and wrote the manuscript.

Competing Interests

C.Y. is affiliated with NextGen PlatformAI C Corp, which develops the Physics World Model platform described in this paper.

- [1] Baker, M. 1,500 scientists lift the lid on reproducibility. *Nature* **533**, 452–454 (2016).
- [2] Wilkinson, M.D. *et al.* The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data* **3**, 160018 (2016).
- [3] Yang, C. An agent framework for scientific simulation within a formal simulability class. *Preprint* (2026). Available at https://github.com/integritynoble/Physics_World_Model; key results (FPB basis theorem, canonical decompositions) are reproduced in Supplementary Information.
- [4] Leuschner, J., Schmidt, M., Baguer, D.O. & Maaß, P. LoDoPaB-CT, a benchmark dataset for low-dose computed tomography reconstruction. *Scientific Data* **8**, 109 (2021).
- [5] Lyu, M. *et al.* M4Raw: a multi-contrast, multi-repetition, multi-channel MRI k-space dataset for low-field MRI research. *Scientific Data* **10**, 264 (2023).
- [6] Meng, Z., Ma, J., & Yuan, X. End-to-end low cost compressive spectral imaging with spatial-spectral self-attention. *Proc. ECCV* (2020).
- [7] Liebgott, H. *et al.* Plane-wave imaging challenge in medical ultrasound. *Proc. IEEE Int. Ultrasonics Symp.* (2016).
- [8] Jannis, D., Hofer, C. & Verbeeck, J. SrTiO₃ [001] 4D-STEM dataset. Zenodo <https://doi.org/10.5281/zenodo.5113449> (2021).
- [9] Barbastathis, G., Ozcan, A. & Situ, G. On the use of deep learning for computational imaging. *Optica* **6**, 921–943 (2019).
- [10] Bertero, M., Boccacci, P. & De Mol, C. *Introduction to Inverse Problems in Imaging* 2nd edn (CRC Press, 2021).
- [11] Arridge, S. *et al.* Solving inverse problems using data-driven models. *Acta Numerica* **28**, 1–174 (2019).
- [12] Lustig, M., Donoho, D. & Pauly, J.M. Sparse MRI: the application of compressed sensing for rapid MR imaging. *Magn. Reson. Med.* **58**, 1182–1195 (2007).
- [13] Candès, E.J., Romberg, J. & Tao, T. Robust uncertainty principles: exact signal reconstruction from highly incomplete frequency information. *IEEE Trans. Inform. Theory* **52**, 489–509 (2006).
- [14] Rudin, L.I., Osher, S. & Fatemi, E. Nonlinear total variation based noise removal algorithms. *Physica D* **60**, 259–268 (1992).
- [15] van Aarle, W. *et al.* The ASTRA Toolbox: a platform for advanced algorithm development in electron tomography. *Ultramicroscopy* **157**, 35–47 (2016).
- [16] Ong, F. & Lustig, M. SigPy: a Python package for high performance iterative reconstruction. *Proc. ISMRM* (2019).
- [17] Adler, J. & Öktem, O. Operator Discretisation Library (ODL). Software available from <https://github.com/odlgroup/odl> (2017).

- [18] Tachella, J. *et al.* DeepInverse: a deep learning framework for inverse problems in imaging. *Preprint at <https://arxiv.org/abs/2312.09828>* (2023).
- [19] Jumper, J. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
- [20] Merchant, A. *et al.* Scaling deep learning for materials discovery. *Nature* **624**, 80–85 (2023).
- [21] Karniadakis, G. E. *et al.* Physics-informed machine learning. *Nature Rev. Phys.* **3**, 422–440 (2021).
- [22] Wang, H. *et al.* Scientific discovery in the age of artificial intelligence. *Nature* **620**, 47–60 (2023).
- [23] Bommasani, R. *et al.* On the opportunities and risks of foundation models. *Preprint at <https://arxiv.org/abs/2108.07258>* (2022).
- [24] Ongie, G. *et al.* Deep learning techniques for inverse problems in imaging. *IEEE J. Sel. Top. Signal Process.* **14**, 171–182 (2020).
- [25] Monga, V., Li, Y. & Eldar, Y. C. Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing. *IEEE Signal Process. Mag.* **38**, 18–44 (2021).
- [26] Kamilov, U. S. *et al.* Plug-and-play methods for integrating physical and learned models in computational imaging. *IEEE Signal Process. Mag.* **40**, 85–97 (2023).
- [27] Aggarwal, H. K., Mani, M. P. & Jacob, M. MoDL: model-based deep learning architecture for inverse problems. *IEEE Trans. Med. Imaging* **38**, 394–405 (2019).
- [28] Adler, J. & Öktem, O. Learned primal-dual reconstruction. *IEEE Trans. Med. Imaging* **37**, 1322–1332 (2018).
- [29] Tian, L., Hunt, B. & Bhatt, M. Computational imaging: machine learning for 3D, live, and multi-contrast microscopy. *Optica* **10**, 464–474 (2023).
- [30] Yuan, X., Brady, D. J. & Katsaggelos, A. K. Snapshot compressive imaging: theory, algorithms, and applications. *IEEE Signal Process. Mag.* **38**, 65–88 (2021).
- [31] Arridge, S. R. & Schotland, J. C. Optical tomography: forward and inverse problems. *Inverse Problems* **25**, 123010 (2009).
- [32] Jiang, Y. *et al.* Electron ptychography of 2D materials to deep sub-ångström resolution. *Nature* **559**, 343–349 (2018).
- [33] Rodenburg, J. M. & Maiden, A. M. Ptychography. In: *Springer Handbook of Microscopy* (Springer, 2019).
- [34] Gustafsson, M. G. L. *et al.* Three-dimensional resolution doubling in wide-field fluorescence microscopy by structured illumination. *Biophysical J.* **94**, 4957–4970 (2008).
- [35] Chambolle, A. & Pock, T. An introduction to continuous optimization for imaging. *Acta Numerica* **25**, 161–319 (2016).
- [36] Beck, A. & Teboulle, M. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM J. Imaging Sci.* **2**, 183–202 (2009).
- [37] Boyd, S., Parikh, N., Chu, E., Peleato, B. & Eckstein, J. Distributed optimization and statistical learning via the alternating direction method of multipliers. *Found. Trends Mach. Learn.* **3**, 1–122 (2011).

- 377 [38] OpenAI. GPT-4 technical report. *Preprint* at <https://arxiv.org/abs/2303.08774> (2023).
- 378 [39] Schick, T. *et al.* Toolformer: language models can teach themselves to use tools. *Proc.*
379 *NeurIPS* (2023).
- 380 [40] Boiko, D. A., MacKnight, R., Kline, B. & Gomes, G. Autonomous chemical research with
381 large language models. *Nature* **624**, 570–578 (2023).
- 382 [41] Pearson, W. R. & Lipman, D. J. Improved tools for biological sequence comparison. *Proc.*
383 *Natl Acad. Sci. USA* **85**, 2444–2448 (1988).
- 384 [42] Kodama, Y., Shumway, M. & Leinonen, R. The Sequence Read Archive: explosive growth
385 of sequencing data. *Nucleic Acids Res.* **40**, D54–D56 (2012).